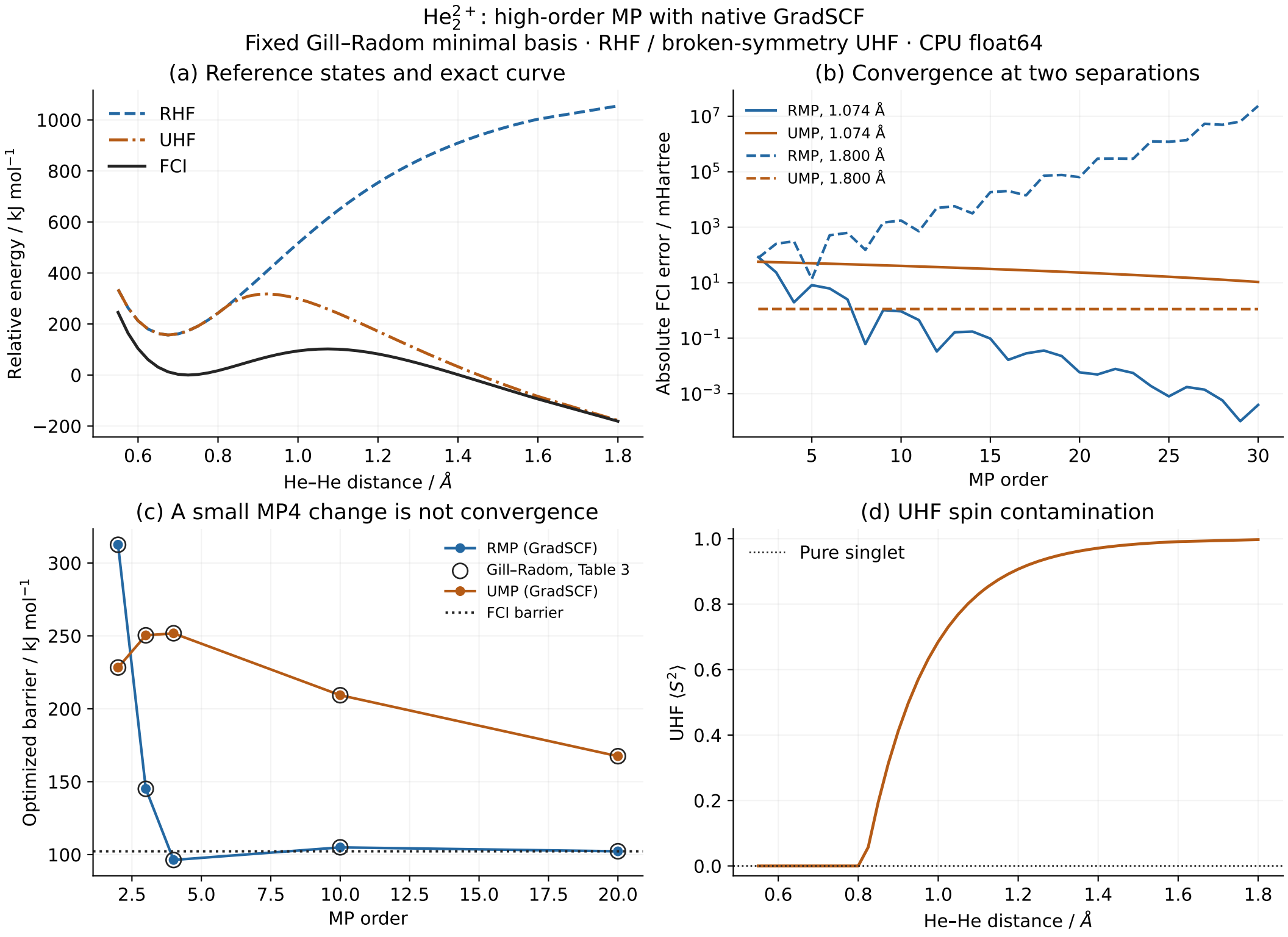
He_2^{2+}: high-order MP with native GradSCF
Fixed Gill–Radom minimal basis · RHF / broken-symmetry UHF · CPU float64
(a) Reference states and exact curve
RHF
UHF
FCI
Relative energy / kJ mol^{-1}
He–He distance / Å
(b) Convergence at two separations
RMP, 1.074 Å
UMP, 1.074 Å
RMP, 1.800 Å
UMP, 1.800 Å
Absolute FCI error / mHartree
MP order
(c) A small MP4 change is not convergence
RMP (GradSCF)
Gill–Radom, Table 3
UMP (GradSCF)
FCI barrier
Optimized barrier / kJ mol^{-1}
MP order
(d) UHF spin contamination
Pure singlet
UHF $\langle S^2 \rangle$
He–He distance / Å